Question 1 (25 marks)

Generate a 3D surface plot of the function $z = 1 - e^{-x^2-y^2}$ for $-1.5 \le x \le 1.5$ and $-1.5 \le y \le 1.5$, with grey color. Overlay contour lines on the surface at $z = 0.2, 0.4, 0.6$ and 0.8. Project the contour lines onto the base plane, as shown in Figure 1. Title the graph as "Projection of $f(x) = 1 - e^{-x^2-y^2}$".

Remark:

1. Adjust transparency of the surface to enhance clarity.
2. Adjust the thickness of the lines to have better appearance.
3. Indicate the projection by dotted-lines.
4. Adjust the angle of view for better appearance.
5. Use *axis square* for the best ratio.
6. Use Latex interpreter in title.

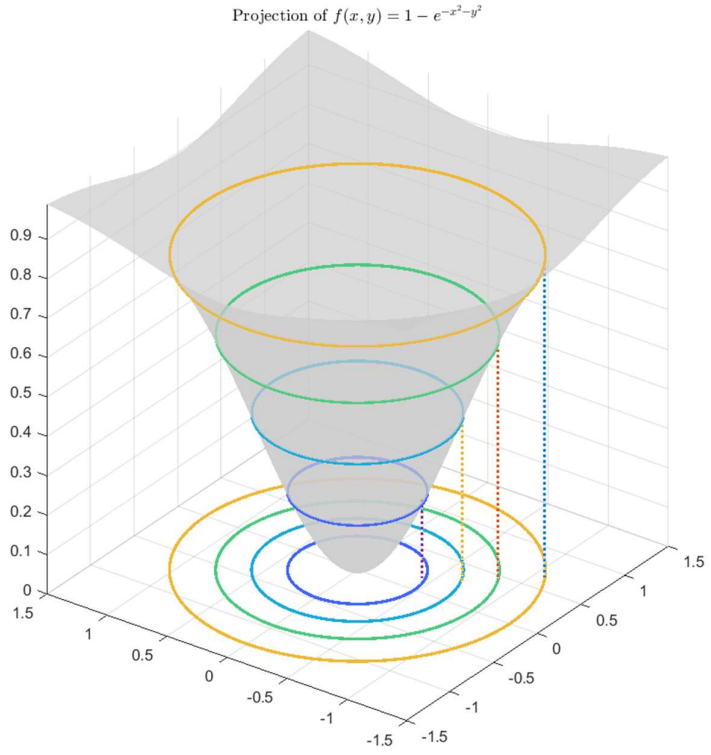


Figure 1

Question 2 (25 marks)

Given a function $f(x,y)=-x^2-y^2$ where $|x|\le 10, |y|\le 10$. The tangent plane of f at the point $(-3,-3,-18)$ is $z=6x+6y+18$. Plot the surface of f and its tangent plane at $(-3,-3,-18)$, together with the lines of $x=-3$ and $y=-3$ on both the surface and tangent plane, as shown in Figure 2. Title the graph as "Intersection of a Surface and Its Tangent Plane" in Cooper Black font.

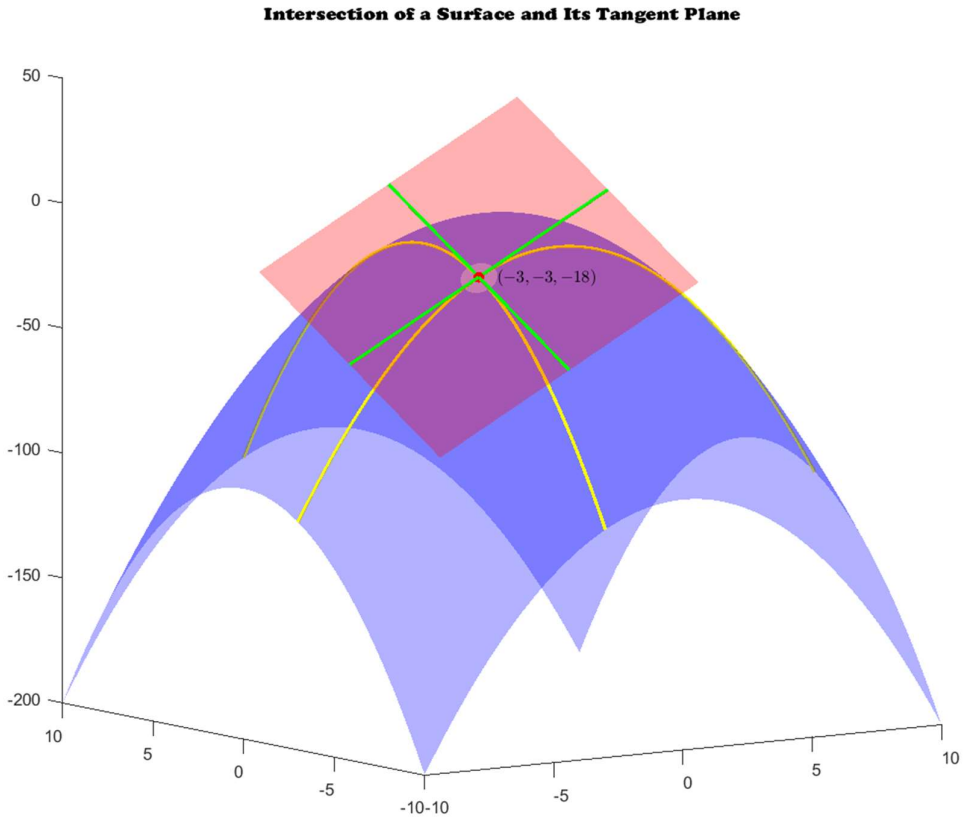


Figure 2